When Do You Apply the Spinor Completeness Relations?
Tracking $u(1)$, $\bar{u}(1)$, $u(4)$, $\bar{u}(4)$ Through $|\mathcal{M}_s|^2$

PHYS 7502

1 The Completeness Relations

Spinor Completeness (Griffiths Eq. 7.99, at $m = 0$)

$$\sum_{s=1,2} u^{(s)}(p)\,\bar{u}^{(s)}(p) = \not{p} + m \xrightarrow{m=0} \not{p} \tag{1}$$

This is a 4×4 **matrix** equation. Written with Dirac indices:

$$\sum_{s=1,2} u_j^{(s)}(p)\,\bar{u}_k^{(s)}(p) = (\not{p})_{jk} \tag{2}$$

When to use it: Whenever you see $u_j(p)$ from one factor sitting next to $\bar{u}_k(p)$ (same particle, same momentum) from another factor, and you are summing over the spin s of that particle. The completeness relation replaces the pair $u_j \bar{u}_k$ with the (j,k) matrix element of $\not{p}$.

2 Where the Spinors Live

Spinor Locations in $\mathcal{M}_s$ and $\mathcal{M}_s^*$

In $\mathcal{M}_s$:

$$\mathcal{M}_s = \frac{-e^2}{s}\,\varepsilon^*_{3\alpha}\varepsilon_{2\beta}\;\underbrace{\bar{u}(4)}_{\text{left end}}\;\gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta\;\underbrace{u(1)}_{\text{right end}} \tag{3}$$

In $\mathcal{M}_s^*$:

$$\mathcal{M}_s^* = \frac{-e^2}{s}\,\varepsilon_{3\rho}\varepsilon^*_{2\sigma}\;\underbrace{\bar{u}(1)}_{\text{left end}}\;\gamma^\sigma(\not{p}_1 + \not{p}_2)\gamma^\rho\;\underbrace{u(4)}_{\text{right end}} \tag{4}$$

Notice: the spinors **swapped positions** under conjugation. $\bar{u}(4)\cdots u(1)$ became $\bar{u}(1)\cdots u(4)$.

3 Stage 1: Write $|\mathcal{M}_s|^2$ with All Spinors Visible

Multiplying $\mathcal{M}_s \times \mathcal{M}_s^*$ (suppressing polarizations and prefactors for clarity):

$$|\mathcal{M}_s|^2 \propto \Big[\bar{u}(4)\,\Gamma_1\,u(1)\Big]\ \Big[\bar{u}(1)\,\Gamma_2\,u(4)\Big] \tag{5}$$

where $\Gamma_1 = \gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta$ (from $\mathcal{M}_s$) and $\Gamma_2 = \gamma^\sigma(\not{p}_1 + \not{p}_2)\gamma^\rho$ (from $\mathcal{M}_s^*$).

Now write out every Dirac index explicitly:

$$\boxed{|\mathcal{M}_s|^2 \propto \bar{u}_i(4)\ (\Gamma_1)_{ij}\ u_j(1)\ \bar{u}_k(1)\ (\Gamma_2)_{k\ell}\ u_\ell(4)} \tag{6}$$

What Are These Indices?

Each spinor has 4 components (the Dirac index runs $i, j, k, \ell = 1, 2, 3, 4$). Each Γ is a 4×4 matrix. The whole expression (6) is a **number** — all indices are contracted (summed over). It is the product of many complex numbers, and numbers commute. This is what lets us rearrange them.

4 Stage 2: Sum Over Particle 1 Spins — FIRST Completeness

This Is When You Apply Completeness for Particle 1

Look at (6). The components $u_j(1)$ and $\bar{u}_k(1)$ are sitting **right next to each other** in the middle of the expression:

$$\bar{u}_i(4)\ (\Gamma_1)_{ij}\quad \underbrace{u_j(1)\ \bar{u}_k(1)}_{\text{apply completeness here}}\quad (\Gamma_2)_{k\ell}\ u_\ell(4)$$

Sum over spins of particle 1:

$$\sum_{s_1} u_j^{(s_1)}(1)\ \bar{u}_k^{(s_1)}(1) = (\not{p}_1)_{jk} \tag{7}$$

Substitute:

$$\sum_{s_1} |\mathcal{M}_s|^2 \propto \bar{u}_i(4)\ (\Gamma_1)_{ij}\ (\not{p}_1)_{jk}\ (\Gamma_2)_{k\ell}\ u_\ell(4) \tag{8}$$

What Just Happened

The pair $u(1)\bar{u}(1)$ got replaced by the 4×4 matrix $\not{p}_1$. The two separate bilinears $[\bar{u}(4)\Gamma_1 u(1)]$ and $[\bar{u}(1)\Gamma_2 u(4)]$ have been **stitched together** into one chain:

$$\bar{u}_i(4)\quad \underbrace{(\Gamma_1\ \not{p}_1\ \Gamma_2)_{i\ell}}_{\text{one big } 4\times4 \text{ matrix}}\quad u_\ell(4)$$

This is now a **single** spinor bilinear $\bar{u}(4)\, Q\, u(4)$, where $Q = \Gamma_1 \not{p}_1 \Gamma_2$.

5 Stage 3: Rearrange to Prepare for Particle 4

We now have:

$$\sum_{s_1} |\mathcal{M}_s|^2 \propto \bar{u}_i(4)\; Q_{i\ell}\; u_\ell(4) \tag{9}$$

This is a number (all indices contracted). Since $\bar{u}_i Q_{i\ell} u_\ell$ is a number, we can write its components in any order. Move $u_\ell(4)$ to the front:

$$= u_\ell(4)\; \bar{u}_i(4)\; Q_{i\ell} \tag{10}$$

Griffiths' "Fancy Footwork" (p. 251)

This step looks illegal but isn't. Each component $u_\ell(4)$, $\bar{u}_i(4)$, $Q_{i\ell}$ is just a complex number. You can write $abc = cab$ for numbers. We moved u_ℓ around to sit next to $\bar{u}_i$ because we **need them adjacent** to apply completeness.
Note the unusual matrix multiplication: $u_\ell \bar{u}_i$ is column $\times$ row $= 4 \times 4$ matrix (the outer product), not row $\times$ column $=$ number.

6 Stage 4: Sum Over Particle 4 Spins — SECOND Completeness

This Is When You Apply Completeness for Particle 4

Now $u_\ell(4)$ and $\bar{u}_i(4)$ are adjacent:

$$\underbrace{u_\ell(4)\; \bar{u}_i(4)}_{\text{apply completeness here}} \qquad Q_{i\ell}$$

Sum over spins of particle 4:

$$\sum_{s_4} u_\ell^{(s_4)}(4)\; \bar{u}_i^{(s_4)}(4) = (\not{p}_4)_{\ell i} \tag{11}$$

Substitute:

$$\sum_{s_4} \sum_{s_1} |\mathcal{M}_s|^2 \propto (\not{p}_4)_{\ell i}\; Q_{i\ell} = (\not{p}_4)_{\ell i}\; (\Gamma_1 \not{p}_1 \Gamma_2)_{i\ell} \tag{12}$$

7 Stage 5: Recognise the Trace

The expression $(\not{p}_4)_{\ell i}\,(\Gamma_1 \not{p}_1 \Gamma_2)_{i\ell}$ is a sum over i and ℓ:

$$\sum_{i,\ell} (\not{p}_4)_{\ell i} (\Gamma_1 \not{p}_1 \Gamma_2)_{i\ell} = \sum_{\ell} \Big[\Gamma_1 \not{p}_1 \Gamma_2 \not{p}_4\Big]_{\ell\ell} \tag{13}$$

But $\sum_\ell M_{\ell\ell}$ is the **definition of the trace**:

Both Completeness Relations → Trace

$$\boxed{\sum_{\text{all spins}} |\mathcal{M}_s|^2 \propto \mathrm{Tr}\Big[\Gamma_1 \not{p}_1 \Gamma_2 \not{p}_4\Big]} \tag{14}$$

Putting in Γ_1 and Γ_2:

$$\boxed{\sum_{\text{all spins}} |\mathcal{M}_s|^2 \propto \mathrm{Tr}\Big[\gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta \not{p}_1 \gamma^\sigma(\not{p}_1 + \not{p}_2)\gamma^\rho \not{p}_4\Big]} \tag{15}$$

No spinors remain. Both u's and both $\bar{u}$'s have been consumed by the two completeness relations, leaving a pure matrix trace.

8 Summary: The Two Completeness Relations in Sequence

Complete Timeline

Start: Four spinors in $|\mathcal{M}_s|^2 = \mathcal{M}_s\mathcal{M}_s^*$:

$$\bar{u}_i(\underbrace{4}_{a})\,(\Gamma_1)_{ij}\,\underbrace{u_j(1)\,\bar{u}_k(1)}_{\text{particle 1 pair}}\,(\Gamma_2)_{k\ell}\,u_\ell(\underbrace{4}_{a})$$

First completeness (sum over particle 1 spins): Replace $u_j(1)\bar{u}_k(1) \to (\not{p}_1)_{jk}$:

$$\bar{u}_i(4)\,\Big[\Gamma_1\,\not{p}_1\,\Gamma_2\Big]_{i\ell}\,u_\ell(4)$$

Two bilinears stitched into one. Three spinors remain ($\bar{u}(4)$, $u(4)$, and $\not{p}_1$).

Rearrange: Move $u_\ell(4)$ to front (it is a number):

$$u_\ell(4)\;\bar{u}_i(4)\,\Big[\Gamma_1\,\not{p}_1\,\Gamma_2\Big]_{i\ell}$$

Second completeness (sum over particle 4 spins): Replace $u_\ell(4)\bar{u}_i(4) \to (\not{p}_4)_{\ell i}$:

$$(\not{p}_4)_{\ell i}\,\Big[\Gamma_1\,\not{p}_1\,\Gamma_2\Big]_{i\ell} = \mathrm{Tr}\Big[\Gamma_1\,\not{p}_1\,\Gamma_2\,\not{p}_4\Big]$$

Loop closed. **Zero spinors remain.** Only a trace of matrices.

9 When Does Each Completeness Get Used?

Answer to "When?"

Order	Which completeness	Trigger
1st	Particle 1: $\sum u(1)\bar{u}(1) = \not{p}_1$	$u(1)$ from $\mathcal{M}_s$ and $\bar{u}(1)$ from $\mathcal{M}_s^*$ are **already adjacent** in the product
2nd	Particle 4: $\sum u(4)\bar{u}(4) = \not{p}_4$	$u(4)$ and $\bar{u}(4)$ become adjacent **after rearranging** (moving $u(4)$ to front)

You **could** do them in either order (particle 4 first, then 1), but the natural order is to do particle 1 first because $u(1)$ and $\bar{u}(1)$ are already adjacent in the product $\mathcal{M}_s\mathcal{M}_s^*$ without any rearrangement.

10 Combined with Photon Sums: Full Ordering

Complete Sequence of Operations

Starting from $|\mathcal{M}_s|^2 = \mathcal{M}_s \times \mathcal{M}_s^*$:

1. **Identify** the four spinors and four polarization vectors in the product.
2. **Spinor completeness #1:** $u(1)\bar{u}(1)$ are adjacent $\rightarrow$ replace with $\not{p}_1$. Two bilinears become one.
3. **Rearrange:** Move $u(4)$ next to $\bar{u}(4)$.
4. **Spinor completeness #2:** $u(4)\bar{u}(4) \rightarrow \not{p}_4$. Everything becomes a trace.
5. **Photon sum #1:** $\varepsilon_3^* \varepsilon_3 \rightarrow -g_{\alpha\rho}$. Contracts indices α with ρ.
6. **Photon sum #2:** $\varepsilon_2 \varepsilon_2^* \rightarrow -g_{\beta\sigma}$. Contracts indices β with σ.

Steps 2–4 and steps 5–6 are **independent** — the spinor sums and photon sums do not interfere with each other. You can do them in any order. The spinor sums eliminate spinors, the photon sums eliminate polarization vectors. After all six steps, you have a pure trace with contracted γ-matrix indices.